Sahoshi

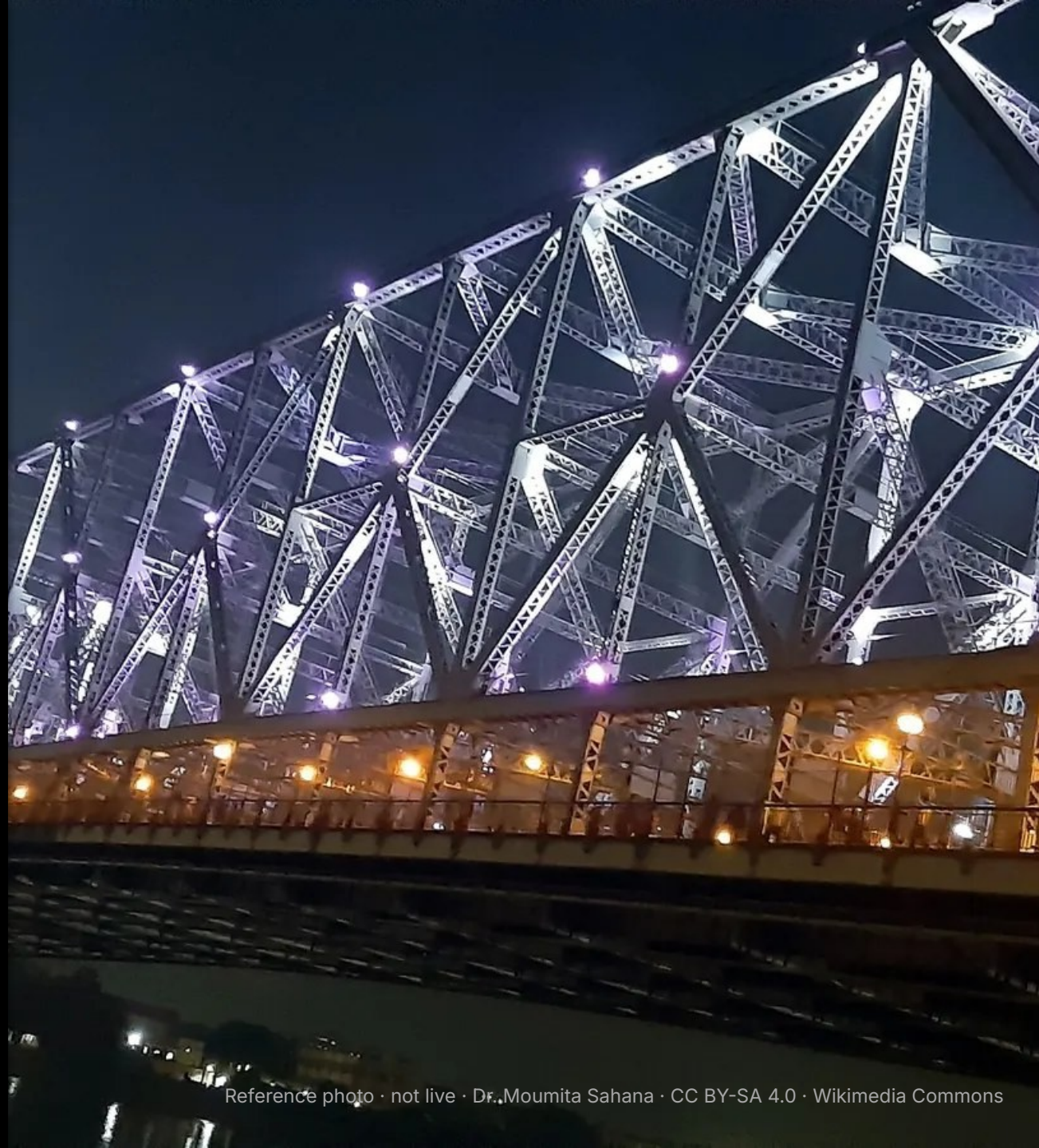
# Not by stopping.

# By steering.

## Keep Kolkata moving.

One live picture of Kolkata's roads, rain and crowds for the traffic police, KMC, disaster response and ambulance (108) crews. The AI proposes; a named officer decides.

Open source (MIT) · runs on one on-premises AI computer · github.com/shubro18202758/sahoshi



Reference photo · not live · Dr. Moumita Sahana · CC BY-SA 4.0 · Wikimedia Commons

THE PROBLEM

# Kolkata loses its roads to the same few shocks, every year

## 251.6 mm

of rain in 24 h on 22–23 Sep 2025 — the heaviest in about four decades; roads went under within the hour, 8–12 people died, most by electrocution in waterlogged streets (IMD; Reuters/PTI).

## High tide

closes the outfall gates on the Hooghly — the same rain waterlogs far faster when it meets a high tide.

## 10,000+

police a day on Durga Puja duty, about 30% on traffic, with ad-hoc one-ways and goods-vehicle bans (Kolkata Police, 2025).

## ≈ 1 lakh

vehicles a day on Howrah Bridge alone; one breakdown cascades across both banks.

## 15–45 min

corridor freezes for VIP movements between Nabanna, the city and the airport.

## One picture missing

traffic police, KMC, WBSDMA, the power utility and ambulance (108) each hold a piece — coordination runs on phone and radio.

# See · Decide · Act — one shared picture

## See

Roads, live traffic, rain, the Hooghly tide, pumps, IMD/NDMA alerts, Bengali and English news, field reports spoken in Bengali, camera road state — on one map.

## Look ahead

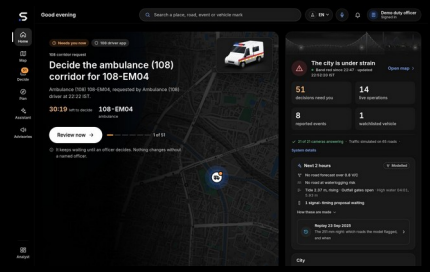
Corridor congestion forecast 15–120 min ahead with uncertainty; a waterlogging nowcast ranks every road by risk with its lead time.

## Decide

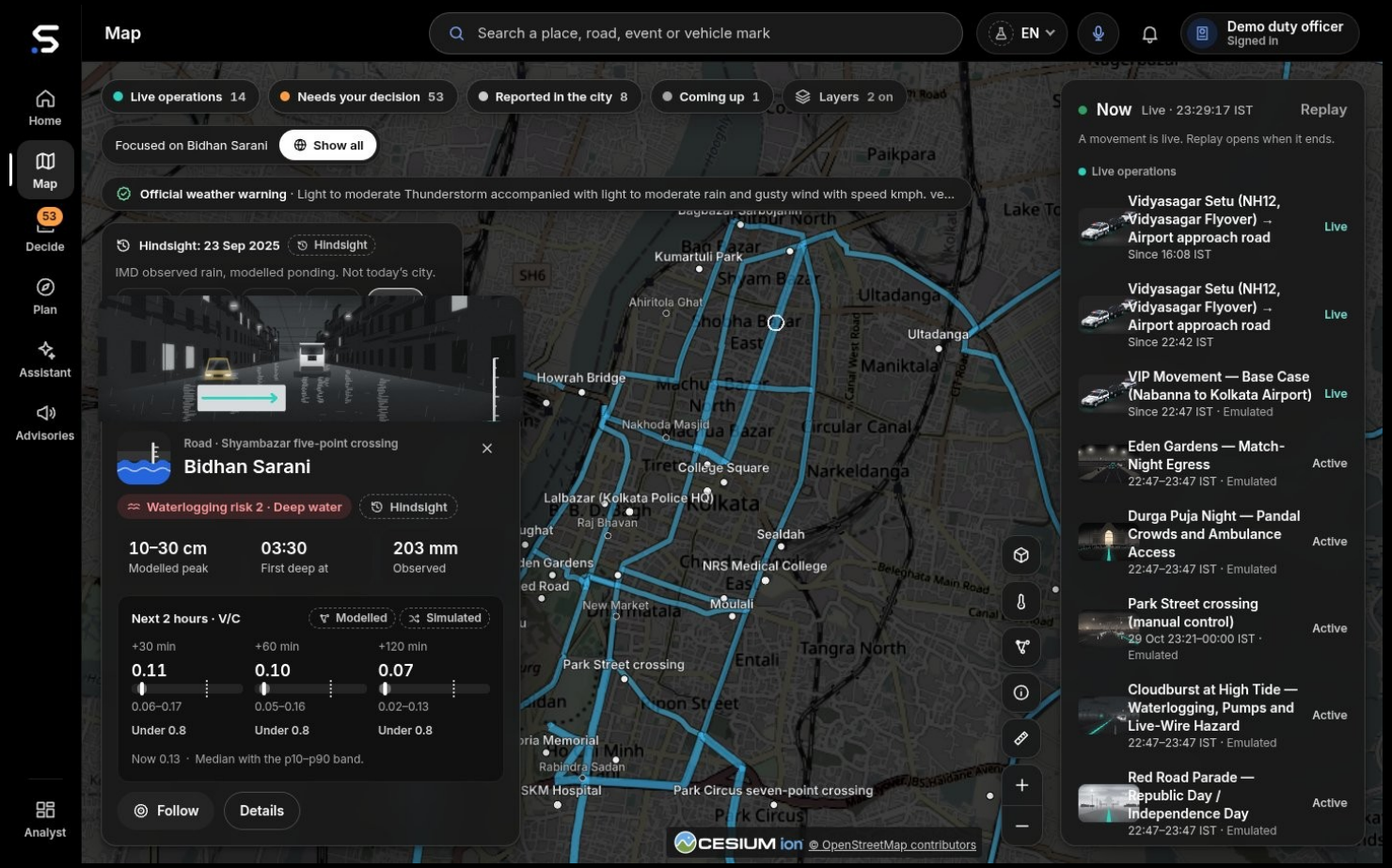
One proposal at a time with what changes if approved: a diversion, a signal-timing change tested in a traffic twin, a rolling hold instead of a freeze, an ambulance window, a live-wire hazard request.

## Act

A named officer approves or rejects — nothing acts on its own; no answer, the plan holds. Advisories go out in English, Bengali and Hindi, read aloud.



# Waterlogging risk and the next two hours, on every road




## Rain × tide × terrain × pumps

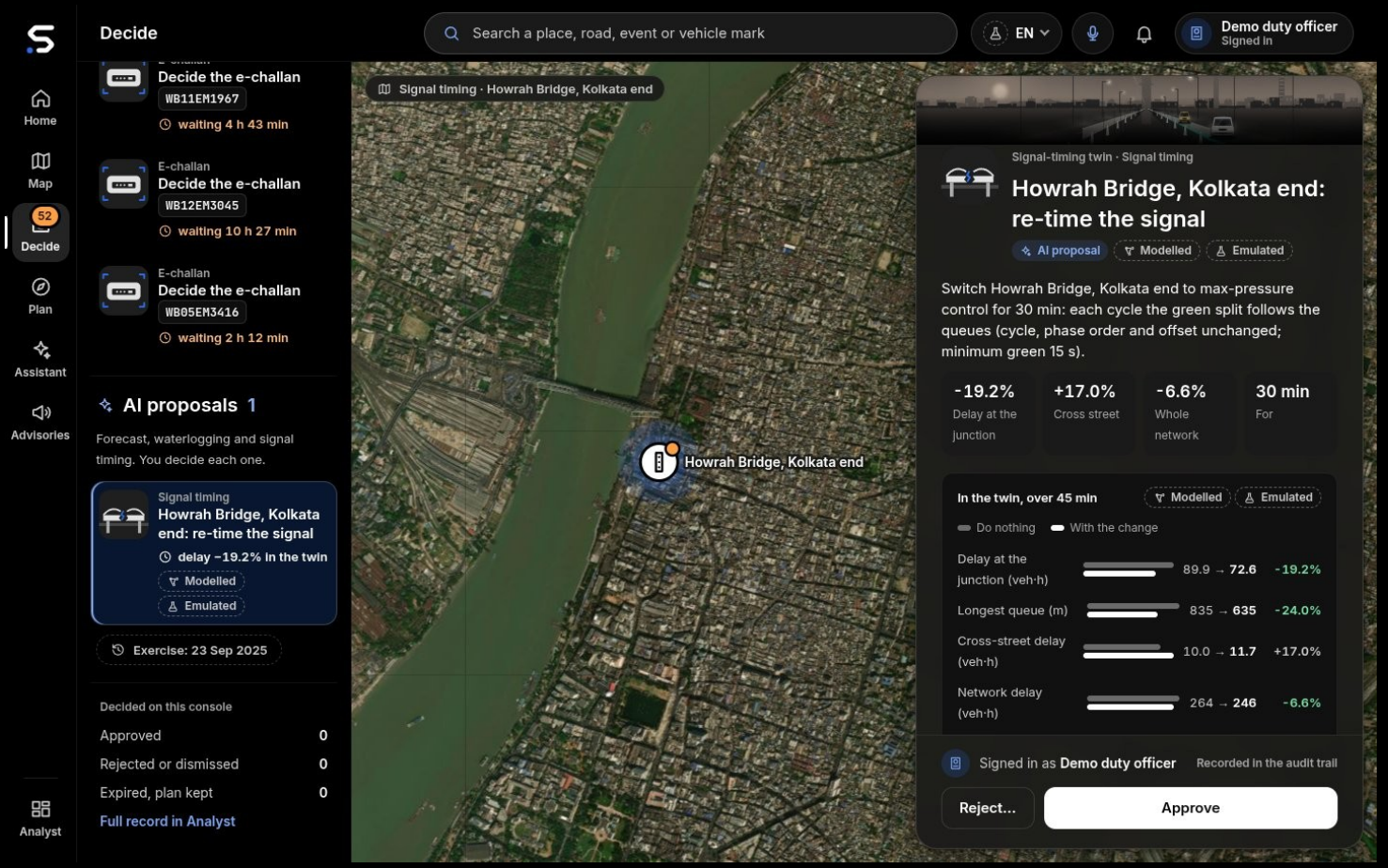
Rain: Open-Meteo, LIVE. Tide: Hooghly model at Garden Reach, MODELLED. Terrain: Copernicus 30 m DEM and OSM drains. Pumps: KMC FFEWS adapter, EMULATED until access.

Risk class 0–3 per road, depth band, lead time to deep water — and the drivers behind it, each labelled.

## Forecast

Chronos-2 forecasts every corridor 15–120 min ahead with p10–p90 bands; crossings of 0.8 V/C become diversion proposals before the queue forms. SIMULATED traffic until Kolkata traffic is recorded.

# The AI proposes. A named officer decides.




## Tested before it is shown

Every signal-timing proposal is simulated in a cell-transmission traffic twin of Kolkata against doing nothing: delay, queues, cross streets, the whole network. It is offered only if it helps without moving the jam.

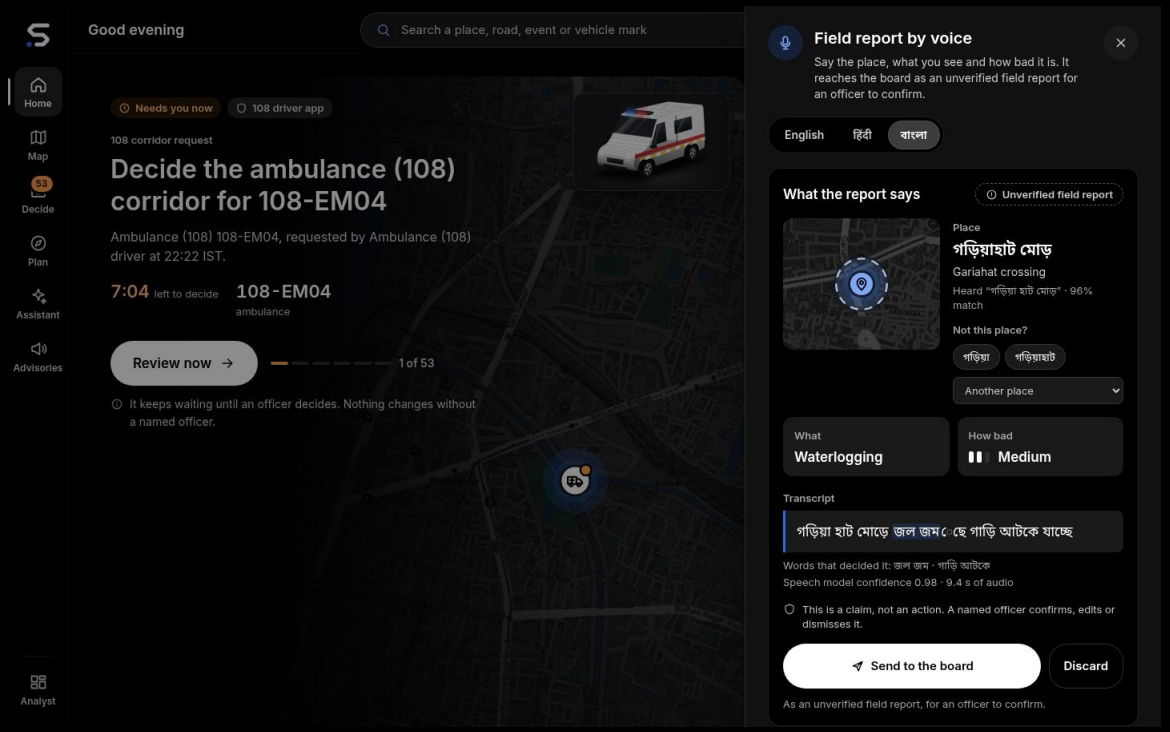
## Accountable

Evidence, what changes, confidence and assumptions on every card. Approve or reject is recorded with the officer's name. No answer: the current plan holds.

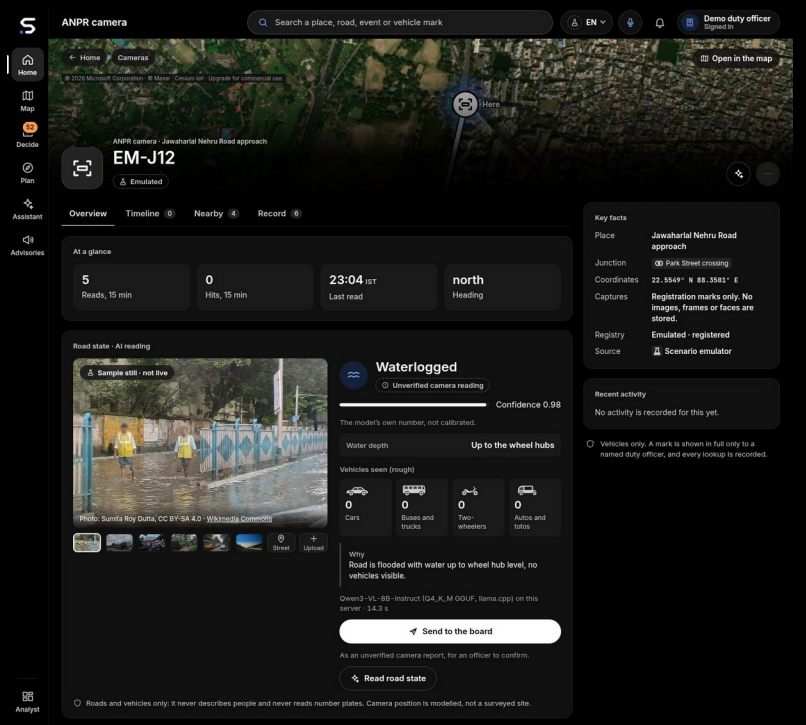
## Explained in two languages

A local language model writes the rationale in English and Bengali; any number not in the twin's metrics is rejected.

# Voice reports and camera road state



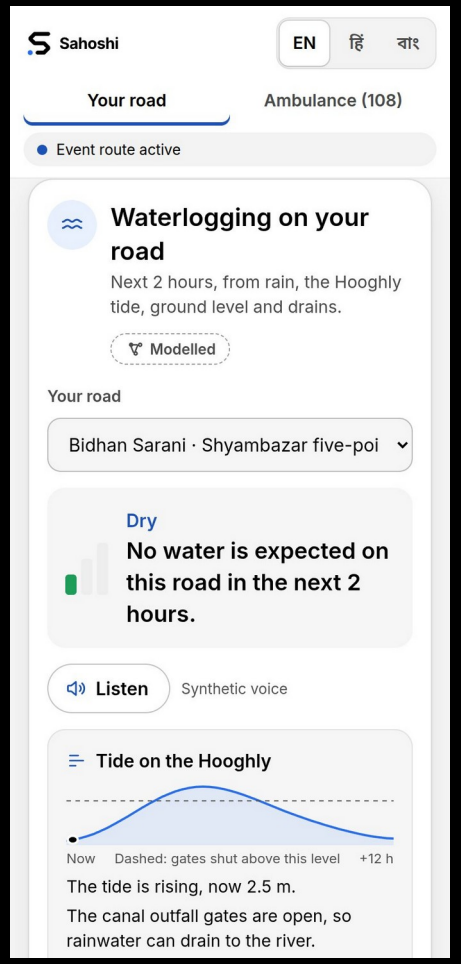

An officer speaks in English, Bengali or Hindi; the on-premises speech model returns place · kind · severity with the transcript — an unverified field report for a named officer to confirm. Audio is not stored.

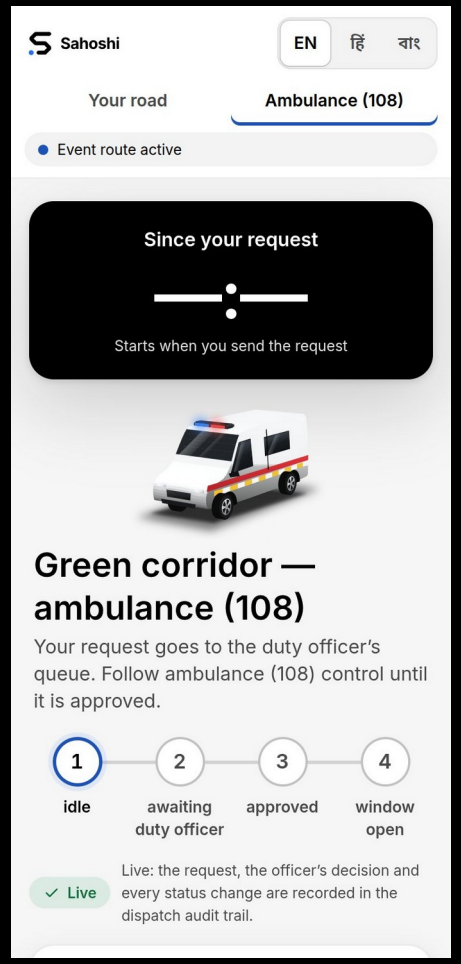



Qwen3-VL reads waterlogged / blocked / tree-fall / stalled bus / clear — roads and vehicles only. It refuses any question about people and never reads plates.

FOR CITIZENS

# Check your road, read aloud




## Plain words, own language

Waterlogging on your road, the tide and the rain, holds and diversions — in English, Bengali and Hindi, with a Listen button (synthetic voice).

## Honest by design

Anything simulated or emulated says so; an officer's unverified report never reaches the public page.

## Ambulance (108) crews

A one-tap green-corridor request that a named duty officer approves.

# 2025–26 open models on one control-room computer

| | | |
|---|---|---|
| **Forecast** | Chronos-2 (Apache-2.0) | Zero-shot corridor forecasts; error 55% below the seasonal baseline in backtest (SIMULATED history) |
| **Waterlogging** | Rain × tide × DEM × pumps | Replay of 22–23 Sep 2025 on IMD-observed rain: 13 of 13 reported places flagged |
| **Signal twin** | Cell-transmission model + Webster / max-pressure | Shockwave speed within 0.2% of theory; a 30-min city run in 0.12 s |
| **Speech** | Bengali Whisper + Indic Parler-TTS (Apache-2.0) | 11.2% word error, place 55/60; voice: places 60/60 (round trip) |
| **Vision** | Qwen3-VL-8B (Apache-2.0) | 11 of 12 road states right; 9 of 9 refusal probes refused |
| **Decision** | pplx-decider-v1.2 (27B) + Gemma 4 | Typed answers in 1.0 s; 22 of 26 labelled Kolkata decisions; an officer approves every action |

No data leaves the control room; it keeps working when the internet fails in a cyclone. No licence fees.

EXPECTED IMPACT · MODELLED

# Delay grows with the square of a freeze — so rolling holds change everything

**100×**

less junction-freeze delay (central case; range 25×–506×)

**357,188 → 3,572**

commuter-minutes per VIP movement on the Nabanna → airport corridor (13 junctions, 29 km)

**≈ 33,000**

golden-hour minutes a year returned to ambulance (108) trips

**45–120 min**

warning before a road goes under (2025 replay, IMD rain)

Model outputs on Kolkata's OpenStreetMap network under stated assumptions (IMPACT_REPORT.md) — not field measurements, and never a claim about lives saved.

GUARDRAILS

# Built to strengthen public institutions — safely

## Vehicles only

No faces, no persons, no tracking of individuals. Number plates stay masked until a named duty officer is on duty.

## Human in charge

The AI only proposes. A named officer approves every action; with no answer, the current plan holds. Every decision is audited.

## Honest labels

LIVE, MODELLED, SIMULATED, EMULATED — on every number and every feed that is not yet connected.

## Events, not people

News and social signals are aggregate, about places and situations; posters are never stored or scored.

## Fair enforcement

No e-challan can be approved against a theft victim; penalties are recommendations a human reviews.

## Data stays home

On-premises, retention limits, in the spirit of the DPDP Act 2023.

IMPLEMENTATION

# A 12-month pilot through one monsoon and one Durga Puja

**Month 0–1** — Read-only data MoU: Kolkata Traffic Police, KMC (FFEWS), WBSDMA, ambulance (108); nodal officers named.

**Month 1–2** — Install the on-premises unit; City Pack finalised for the pilot division; adapters switched from EMULATED to LIVE.

**Month 2–4** — Shadow mode: Sahoshi proposes, officers work as usual; weekly comparison.

**Month 5–12** — Live pilot (e.g. Esplanade – Park Street – Rashbehari – Gariahat) through the monsoon and Puja; officers approve.

**Month 12** — Independent review; city-wide roll-out over the next year.

## Cost

**Pilot, 12 months: ≈ ₹45–55 lakh**

Two on-premises AI units ₹8–10 lakh · engineering ₹29 lakh · optional sensors ₹4 lakh · training ₹2 lakh · 15% contingency.

**City-wide, two years: ≈ ₹2–2.5 crore**

Running cost ≈ ₹40–60 lakh a year.

**Software licences: ₹0**

Open models, OpenStreetMap, Sahoshi under MIT — existing CCTV, control rooms and FFEWS sensors are reused.

SCALABILITY

# A city is a folder

The engine — live picture, forecasting, nowcast, voice, vision, decision loop, console — is the same everywhere. A City Pack holds what differs: the OpenStreetMap extract and corridors, gazetteer, agencies and emergency numbers, plate format, UI languages, local scenarios and impact parameters.

The prototype proves it: it was first built for another Indian city, and the same build tools now produce Kolkata; a test builds both.

Per district: about 1–2 weeks of local data work, local adapters (CCTV, signals, flood sensors) and local SOPs. The software, models and hardware stay the same.

**Howrah · Bidhannagar**

already inside the Kolkata pack

**Siliguri**

gateway to the North-East; NH10/NH31 choke-points

**Asansol–Durgapur**

industrial corridor

**Patna · Ranchi**

Bihar and Jharkhand; Hindi already first-class

EVIDENCE

# It runs today

## Working prototype

Console, citizen and driver pages, five AI services and the traffic twin, on one on-premises unit.

## Real Kolkata roads

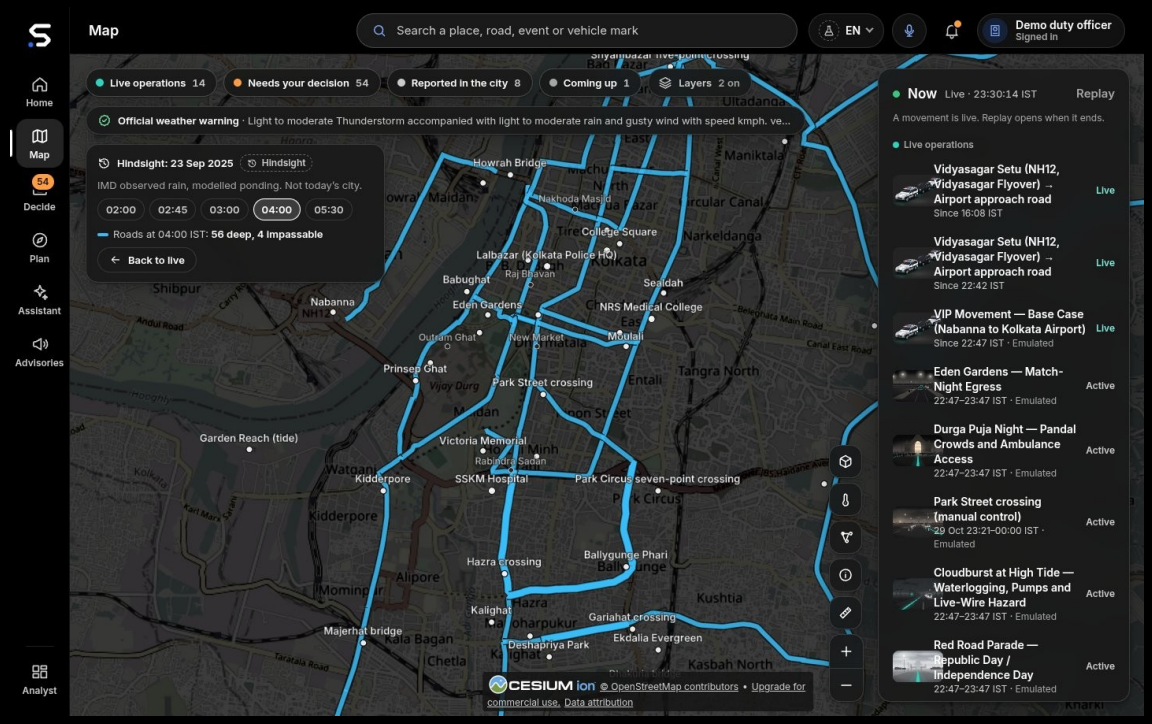
49 junctions, 65 road segments, 125 km from OpenStreetMap; the Nabanna → airport route is 29 km.

## Thousands of automated tests

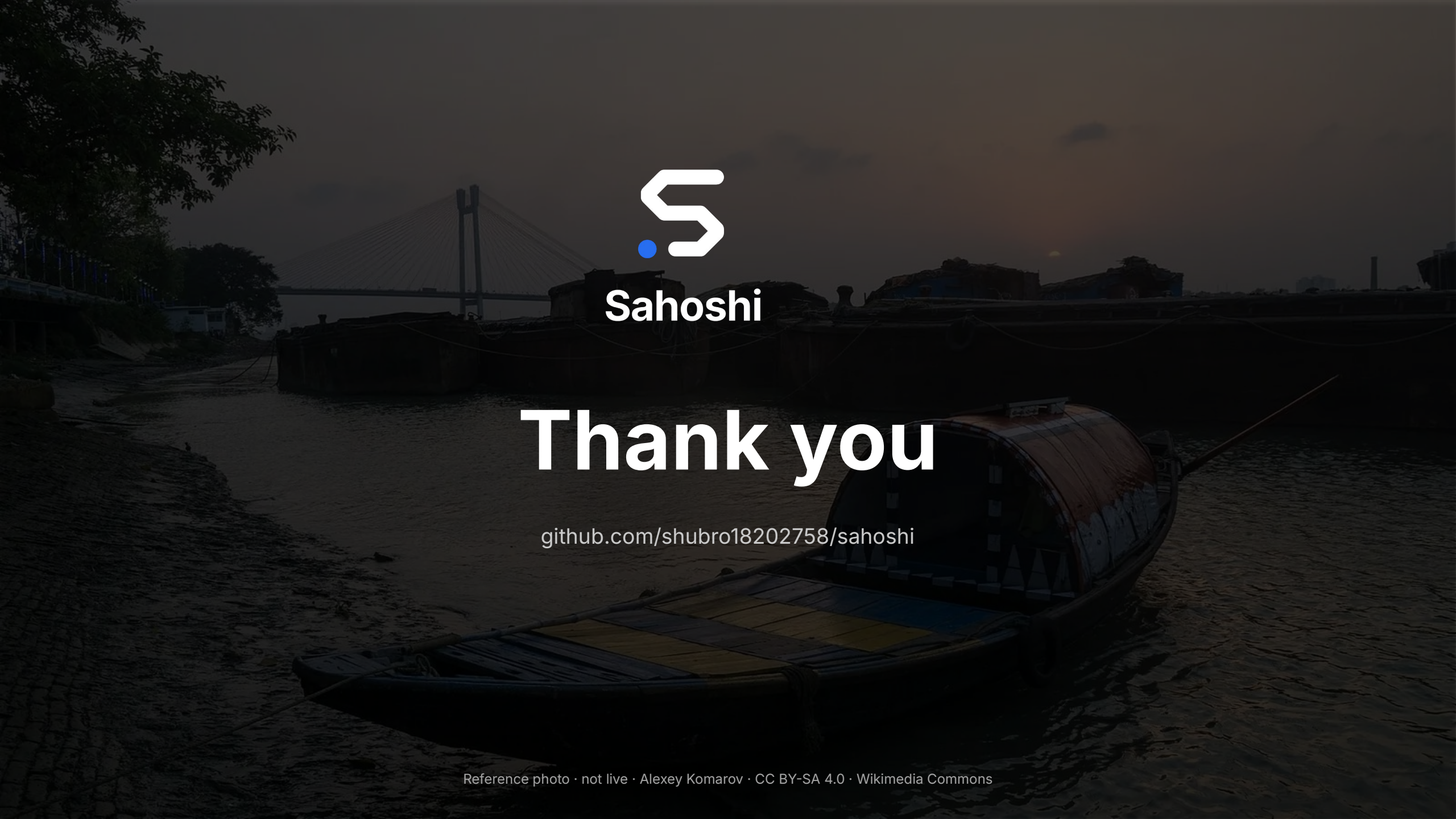
Python suites, the console's unit and browser tests, contract replays and Rust tests — run on every change.

## Open source

github.com/shubro18202758/sahoshi — setup, technical design, sample data and test cases.




Replay of 22–23 Sep 2025 on IMD-observed rain (OBSERVED rain, MODELLED ponding): every news-reported waterlogged place on the network was flagged — the night almost the whole city flooded.



Sahoshi

# Thank you

github.com/shubro18202758/sahoshi

Reference photo · not live · Alexey Komarov · CC BY-SA 4.0 · Wikimedia Commons